Md Kamal Uddin

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RESEARCH INTERESTS

- **Condensed Matter Physics**
- **Nanomaterials and Functional Materials**
- **Nonlinear Optics**
- **Photocatalysis and Environmental Nanotechnology**
- **Optical Characterization of Materials**

EDUCATION

University of Dhaka, Bangladesh

M.S. in Physics: CGPA: 3.82/4.00; Dean's Honor Award

University of Dhaka, Bangladesh

B.S. in Physics: CGPA: 3.85/4.00; Ranked **2nd among 140 students**

PROFESSIONAL EXPERIENCE

Lecturer | Department of Physics, University of Dhaka — January 2023 – Present

Lecturer | Department of Physics, Bangladesh University of Engineering and Technology — 2022

Graduate Teaching Assistant | **BRAC University**, Dhaka — 2022

RESEARCH EXPERIENCE

- **Functional Nanomaterials for Environmental Applications**

  Research on synthesis and characterization of graphene oxide-based semiconductor nanocomposites for photocatalytic degradation of organic pollutants. Techniques used: XRD|FESEM |Raman Spectroscopy |FTIR |UV-Visible |Spectroscopy | Photoluminescence | EDX | Rietveld Refinemen

- **Nonlinear Optical Materials**

  Research involving: Z-scan technique |Thermal lens measurements |Third-order nonlinear optical properties |Optical limiting behavior

- **Structural and Electronic Properties**

  Experience with: Density Functional Theory (DFT) |Crystal structure refinement |Band gap engineering |Defect analysis

TECHNICAL SKILLS

OriginPro | MATLAB | Python | C++ | VASP | VESTA | FullProf Suite | GSAS-II | LaTeX

AWARDS AND HONORS

- Dean's Honor Award
- National Science and Technology Fellow
- Semiconductor Technology Research Centre Fellow
- National Talent Pool scholarship

PUBLICATIONS

1. P. D. Gupta, M. K. Uddin et al., “Tuning the nonlinear optical properties of sno 2–rgo nanocomposites: Exploration using conventional z-scan and thermal lensing models,” Materials Advances, 2025. https://DOI:10.1039/d5ma00790a . (Featured as cover)

2. M. K. Uddin, N. Deb, R. Rashid, H. Das, I. M. Syed, and S. M. Hoque, “Physical properties with high specific loss power of magnetite (fe3o4) synthesized via thermal decomposition technique,” AIP Advances, vol. 13, no. 10, 2023. https://doi.org/10.1063/5.0164802.

3. S. Jahan, M. A. Sayem, M. K. Uddin, F. Ferdous, R. A. Jahan, and I. M. Syed, “Remarkable photocat alytic performance and recyclable pathway of cleansing wastewater using zno–rgo nanocomposite,” AIP Advances, vol. 15, no. 7, 2025. https://doi.org/10.1063/5.0275565

4. U.S.A. Meem, R.A. Jahan, M.K. Uddin, *et al.* Functional development and enhanced photocatalytic efficiency of GO-supported SnO_2 by optimizing composition. *J Mater. Sci: Mater Eng.* **21**, 34 (2026). https://doi.org/10.1186/s40712-026-00403-2

CURRENT PROJECTS

1. Enhancement of optical limiting in $Zn_{1-x}Nd_xO$-GO using Nd.

2. Antibiotic removal from waste water using Fe3O4-GO Nanocomposites.

3. Nonlinear Properties of V2O5-GO using Z-scan thechnique.

REFERENCES

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